

7COM1069 – Cyber Operations

Topic: Situational Awareness

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1. Executive Summary

The Thames Water Treatment Facility or WTCS experienced a coordinated and multi-layered cyberattack that combined network intrusion, operational technology interference, and a sophisticated psychological disinformation campaign. This hybrid operation targeted the United Kingdom's critical national infrastructure with the objective of undermining water safety, eroding public trust, and destabilising regional social and economic stability.

Technical evidence recovered from the network packet capture (PCAP) confirms that attackers gained initial access through stolen administrator credentials obtained via a spear-phishing campaign. Once inside the corporate network, the adversary conducted reconnaissance, moved laterally across internal systems, and pivoted into the industrial control system environment. Command-and-control (C2) traffic was established with an external server, enabling remote attacker interaction. Packet analysis further indicates targeted probing of Programmable Logic Controllers (PLCs) and potential attempts to manipulate chemical dosing parameters – an action that could compromise water purification processes. Although full operational sabotage was not successful, the attempt itself represents a severe breach of national infrastructure security.

Simultaneously, the attackers launched an orchestrated psychological operation across social media, messaging platform, fabricated news sites, manipulated CCTV footages, and deepfake broadcast materials. This campaign falsely claimed widespread contamination of London's water supply with hazardous chemicals. This misinformation spread rapidly, creating public panic, overwhelming local authorities with enquiries, and driving residents to hoard bottled water. The convergence of genuine ICS interference with engineered public fear amplified the perceived severity of the incident far beyond the actual technical impact.

The combined attack led to significant implications, a temporary loss of confidence in water safety, operational disruption within the treatment facility, economic pressure on supply chains, and increased public anxiety. From a national security standpoint, the incident demonstrated how cyber-physical environments and information ecosystems can be exploited simultaneously to generate strategic instability.

This report provided a detailed technical analysis of the cyber components, and assessment of the psychological warfare elements, and a unified defense strategy to mitigate future hybrid threats targeting the nation's critical infrastructure.

2. Technical Analysis of Cyber Components

2.1. Methodology

The PCAP file `Thames_water_attack.pcap` was analysed using Wireshark/tshark and Zeek to reconstruct the attack lifecycle. Filtering focused on:

- Internal ICS subnet traffic (10.0.0.0/24 assumed).
- Ports commonly abused in ICS and Windows environments (502/Modbus, 445/SMB, 3389/RDP, 22/SSH, 80/443/HTTP(S)).
- External connections indicative of C2 or exfiltration.

Traffic was then mapped to the phases of the intrusion: reconnaissance, exploitation, command and control, and exfiltration/sabotage.

2.2. Reconnaissance

During the early phase, the attacker operating with stolen administrator credentials from an internal host conducts internal network discovery and service enumeration.

What you should see in the PCAP?

- A single internal source IP sending many TCP SYNs to multiple destination ports on several internal hosts.
- Probing of 445(SMB), 3389 (RDP), 22 (SSH) from IT side and 502 (Modbus), 80/443 (HTTP) from ICS side.

This pattern is consistent with an authenticated attacker mapping the flat OT/IT network to identify accessible services and potential pivot points.

Commands to run:

- a) Basic conversation overview
`tshark -r Thames_water_attack.pcap -q -z conv,ip`
- b) Identify SYN scan pattern
`tshark -r Thames_water_attack.pcap \ -Y "tcp.flags.syn==1 && tcp.flags.ack==0" \ -T fields -e ip.src -e ip.dst -e tcp.dstport | sort | uniq -c | head`
- c) Focus on ICS-related ports
`tshark -r Thames_water_attack.pcap \ -Y "tcp.dstport==502 || tcp.dstport==44818 || tcp.dstport==20000" \ -T fields -e frame.time -e ip.src -e ip.dst -e tcp.dstport`

2.3. Exploitation

Once potential targets are discovered, the attacker attempts to exploit vulnerable services to deepen access and reach control systems.

Likely focus areas:

- SMB/Windows exploitation against file shares/admin systems.
- Web admin interface abuse on ICM HMIs over HTTP/HTTPS
- Modbus write attempts to controllers.

a) SMB/Windows exploitation

SMB2 `SESSION_SETUP` AND `TREE_CONNECT` messages from host 10.0.0.X to 10.0.0.Y indicates that attacker leveraging compromised credentials to access file shares and potentially obtain SCADA/PLC project files.

Command:

```
tshark -r Thames_water_attack.pcap \ -Y "tcp.port==445 && smb2" \ -T fields -e frame.time -e ip.src -e ip.dst -e smb2.cmd | head -n 30
```

b) Web service exploitation (ICS admin panels)

Repeated HTTP(S) requestes from the compromised workstation to the ICS management interface at 10.0.0.51 suggest attempts to exploit a vulnerable web configuration panel, likely targeting weak authentication or known ICS CVEs.

Command:

```
tshark -r Thames_water_attack.pcap \ -Y "(ip.dst==10.0.0.30 || ip.dst==10.0.0.51) && (tcp.port==80 || tcp.port==443)" \ -T fields -e frame.time -e ip.src -e ip.dst -e http.request.method -e http.request.uri 2>/dev/null | head -n 20
```

c) PLC/Modbus exploitation

Modbus function codes associated with write operations were observed from the attacker-controlled host towards the PLC at 10.0.0.30, indicating an attempt to change process parameters or disable alarms. Even though change verification later showed no persistent modification, these packets demonstrate an explicit sabotage attempt.

Command:

```
tshark -r Thames_water_attack.pcap \ -Y "tcp.port==502" \ -T fields -e frame.time -e ip.src -e ip.dst -e modbus.func_code -e modbus.reference_num -e modbus.data | head -n 30
```

2.4. Command & Control

C2 is where the compromised host phones home to a remote server for instructions. A persistent TCP conversation between internal host 10.0.0.X and external address 203.0.113.5 with regular low-volume packets indicates beaconing behaviour typical of a Remote Access Trojan (RAT). The frequency and symmetry of payloads suggest command polling rather than bulk transfer.

Commands:

a) Find the main external IPs:

```
tshark -r Thames_water_attack.pcap \ -Y "ip && !ip.dst==10.0.0.0/24 && !ip.src==10.0.0.0/24" \ -T fields -e ip.src -e ip.dst | sort | uniq -c | sort -nr | head
```

b) Zoom into suspected C2:

```
tshark -r Thames_water_attack.pcap \ -Y "ip.addr==203.0.113.5" \ -T fields -e frame.time_relative -e ip.src -e ip.dst -e tcp.len | head -n 40
```

c) Show TCP conversation summary:

```
tshark -r Thames_water_attack.pcap -q \ -z conv,tcp,"ip.addr==203.0.113.5"
```

2.5. Exfiltration or Sabotage

Analysis of TCP port 502 traffic shows targeted attempts to write to PLC registers, which indicates intended sabotage of water treatment parameters. However, conversation and I/O statistics show no sustained high-volume outbound transfers, suggesting the attack was focused on disruption and public panic, not large-scale data exfiltration.

Commands:

a) Global I/O statistics

```
tshark -r Thames_water_attack.pcap -q -z io,stat,60
```

b) Big external conversations

```
tshark -r Thames_water_attack.pcap -q \ -z conv,ip
```

2.6. Vulnerabilities Targeted

1. Credential Compromise

- Stolen administrator credential via spear-phishing allowed legitimate remote access.
- No multi-factor authentication or conditional access in place.

2. Flat/Poorly Segmented Network

- Same admin credentials worked across IT and OT.
- Recon and exploitation traffic flowed freely between hosts and corporate workstations.

3. Unsecured ICS Protocols

- Modbus (TCP/502) used without authentication or encryption.
- Attacker able to send controller commands directly.

4. Exposed/Weak Web Admin Interfaces

- ICS controller/HMI accessible over HTTP(S) from the IT network.

5. Insufficient Monitoring/Detection

- Port scanning, SMB lateral movements, and C2 beaconing were not automatically blocked.
- No specific ICS-aware IDS signature for Modbus write attempts.

3. Impact Assessment

The cyberattack against the Thames Water Treatment Facility (WTCS), combined with a coordinated psychological disinformation campaign, produced operational, public health, social, economic, and national security consequences. This section evaluates both the actual impact observed during the incident and the potential impact the attack had succeeded fully.

3.1. Operational Impact on the Water Treatment Facility

The cyber intrusion directly targeted the industrial control system responsible for water purification, chemical dosing, and environmental monitoring. Analysis of the PCAP indicates attempted manipulation of Programmable Logic Controllers (PLCs) using Modbus-like write operations and unauthorized communication with dosing-control panels.

Actual Operational Impact:

- Temporary instability in ICS communication.
- Short-term disruption to monitoring capabilities.
- Increased workload for staff.

Potential Operational Impact

If the attacker successfully manipulated dosing controllers or disabled safety interlocks, consequences could include:

- Under-chlorination
- Over-chlorination
- False sensor readings
- Forced shutdown

In worst-case scenario, deliberate manipulation could result in a mass public health emergency requiring significant medical, logistical, and political intervention.

3.2. Public Health Impact

Although no chemical were actually altered, the attack generated perceived contamination, which has its own health implications.

Health-related consequences included

- Psychosomatic symptoms
- Increased pressure on emergency services
- Water avoidance behaviour
- Risk of secondary health issues

The absence of physical contamination did not prevent a public health burden caused by fear and misinformation.

3.3. Social Impact

The psychological operation had immediate and widespread societal consequences

a) Loss of public trust

Widespread distribution of fake news articles, manipulated CCTV footage, deepfake BBC broadcasts, and fabricated group chats caused many citizens to distrust.

b) Public panic and behavioural disruption.

Supermarkets experienced panic buying of bottled water, leading to shortage. Schools and businesses temporarily suspended tap water usage.

c) Breakdown of social cohesion

Misinformation campaigns intentionally divided the public, targeting: Parents, residents of specific boroughs, and vulnerable communities.

3.4. Economic impact

Although the technical damage was limited, the combined effects of panic, emergency operations, and misinformation carried significant economic consequences.

- a) Direct cost: Incident response overhead, Infrastructure inspections, and temporary operational slowdown.
- b) Indirect costs: Supply chain disruption, loss of productivity, and consumer behaviour changes.
- c) Long-term financial impact: Loss of trust in water safety, and even combined attack imposes a lasting economic shadow on critical infrastructure.

3.5. National Security Impact

This attack demonstrated the vulnerability of the UK's critical national infrastructure to hybrid cyber-psychological warfare, the implications extend far beyond the Thames Water region.

a) Strategic-level effects:

Erosion of public confidence in the government's ability to protect essential services.

Exploitation by hostile states to test response capability and gather intelligences on CNI weaknesses.

b) Amplification via psychological operations

The attacker's deepfake media campaign achieved exactly what modern hostile actors aim for:

Undermining truth

Distorting information ecosystems

Creating chaos during technical incidents

Turning small disruption into public crises

c) Combined impact

Individually, the technical breach and misinformation campaign would be manageable. Together, they delivered strategic impact disproportionate to the attacker's technical success.

4. Threat Intelligence

4.1. Attacker Profile (MOC – Motivation, Opportunity, Capability)

Based on analysis of the PCAP data and the supplied multimedia evidence, the attack can be attributed to a highly capable, strategically motivated actor, consistent with a state-sponsored or state-aligned APT rather than a purely criminal group.

a) Motivation:

The operation clearly prioritized disruption and destabilization over direct financial gain:

- The primary technical focus was on ICS access and potential sabotage of chemical dosing, not on data theft or ransom.
- Simultaneous deployment of disinformation, deepfakes, and fake news indicates an intent to:
 - Undermine trust in public utilities and government.
 - Generate social unrest and panic.
 - Expose or test the resilience of UK Critical National Infrastructure.

This aligns with the strategic objectives of hostile states engaged in hybrid warfare, where cyber operations and information operations are combined to weaken an adversary without open conflict.

b) Opportunity

The adversary exploited several concrete opportunities:

- Compromised administrator credentials, giving them direct entry to the internal network.
- Flat or weakly segmented OT/IT environment, allowing lateral movement from corporate systems into the ICS network.
- Unprotected ICS protocols and exposed management interfaces on controllers, enabling probing and attempted manipulation of dosing equipment.
- Lack of robust public communication strategy, allowing misinformation to spread faster than official updates.

The combination of poor identity controls, inadequate segmentation, and limited ICS-specific monitoring greatly increased the attacker's opportunity to operate undetected in both cyberspace and the information space.

c) Capability

The attacker's capability is evident in both technical and psychological dimensions:

- Technical capability:
 - Use of C2 infrastructure and consistent beaconing to a single external IP.
 - Knowledge of ICS protocols and targeting of specific PLC/dosing controller endpoints
 - Execution of lateral movement across internal Windows/SMB environments.
- Information operations capability:
 - Production of fake BBC-style video, doctored CCTV, and plausible but fabricated news articles.
 - Seeding of fake group chat screenshots and social media posts designed to mimic genuine local conversations.

The sophistication of the technical intrusion and the psychological operation strongly suggest a well-resourced, organized threat actor with access to specialist skills in ICS, media manipulation, and the influence operations.

5. Disinformation Campaign Analysis

The disinformation component functioned as a psychological weapon designed to multiply the effect of the technical intrusion.

5.1. Deepfakes and Morphed Media – Content and Target Audience

The supplied multimedia includes:

- Deepfake-style news video mimicking a BBC broadcast.
- Doctored CCTV footage from within the treatment facility.
- Fake online news articles and newspaper front pages.
- Fabricated group chats and social media posts.

Target Audience

The campaign was clearly aimed at:

- Residents of London and surrounding areas supplied by Thames Water.
- Parents and families, using emotive language about “children drinking toxic water”.
- Social-media-active demographics, via Twitter-style posts and WhatsApp-style chats.
- Politically sensitive individuals, by suggesting government cover-ups and negligence.

The psychological design is simple but effective: strike at something people cannot opt out of (water) and weaponise fear of contamination.

Fake screenshots and articles amplified this with phrases implying systemic failure, “cover-up”, and “emergency contamination warning suppressed”. The attacker’s aim was to delegitimise official information sources while promoting their own false narrative.

5.2. Impact on Public Perception and Behaviour

The disinformation campaign had several observable and predictable effects:

- Panic and anxiety: People believed the water was unsafe even after official reassurances.
- Behavioural change: A sharp increase in bottled water purchases, avoidance of tap water, and hoarding behaviour.
- Increased distrust: Residents questioned statements from Thames Water, local councils, and national government, viewing them as complicit or incompetent.
- Information overload: The sheer volume of conflicting information made it difficult for individuals to distinguish truth from fabrication.

Even in the absence of actual contamination, the perceived threat materially altered public behaviour and strained emergency communication systems.

5.3. Distribution Methods of Deepfake Media

The distribution showed a classic multi-channel amplification strategy:

a) Social Media Platforms (X/Twitter, Facebook, Instagram)

- Short clips of the fake BBC video.
- Eye-catching headlines and edited screenshots of fake articles.
- Hashtags like #ThamesWaterToxic, #LondonWaterCrisis.

b) Messaging Apps (WhatsApp/Telegram)

- Circulation of group chat screenshots and forwarded voice notes.
- Links to fake “official PDFs” purporting to be government or NHS guidance.

- Chain messages urging people to “share quickly before it’s taken down”.

c) Fake News Websites / Cloned Branding

- News pages mimicking BBC/Guardian design with minor visual distortions.
- URLs closely resembling legitimate news domains (e.g., typosquatting).
- Shared via social media posts as “proof”.

d) Localised Micro-targeting

- References to specific London boroughs and landmarks (“water in East London is unsafe”, “Hounslow hospital overwhelmed”), making the threat feel personal and local.

This distribution pattern is designed for virality and resilience – even if one platform reacts and removes content, the narrative persists in chats, screenshots, and re-uploads.

6. Defence and Counter-Disinformation Strategy

Defending against a hybrid attack on CNI requires a unified strategy covering technical controls, threat modelling, information operations, and coordinated incident response.

6.1. Threat Model for the Water Treatment Infrastructure

A high-level threat model for Thames WTCS should identify:

Key Assets

a) OT/ICS Assets

- PLCs and dosing controllers.
- SCADA servers and HMIs.
- Historian databases and process logs.

b) IT Assets

- AD domain controllers.
- Email system.
- Engineering workstations and laptops.

c) Information Assets

- Configuration files, dosing recipes.
- Operational manuals and network diagrams.
- Public communication channels and brand reputation.

Adversaries

- State-sponsored or state-aligned APTs.
- Terrorist or extremist groups seeking disruption.
- Highly skilled cybercriminals with access to ICS tooling.

Attack Vectors

- Spear-phishing against administrators and engineers.
- Compromised VPN or remote access portals.
- Web interface vulnerabilities in ICS management consoles.
- Exploitation of unauthenticated protocols (Modbus).
- Social media and messaging platforms used to spread disinformation.

Trust Boundaries

- Boundary between internet and IT networks.
- Boundary between IT and OT (ICS) networks.
- Boundary between OT and safety systems.
- Boundary between internal communications and public messaging.

The threat model should prioritise attacks that cross these boundaries: stolen credentials bridging identity boundaries, and misinformation bridging the gap between technical reality and public perception.

6.2. Technical Security Measures for the Water Treatment System

a) Network Segmentation and Zero Trust

- Strict segregation of IT and OT networks using firewalls and industrial DMZs.
- OT devices should never initiate connections to the internet.
- Access to ICS from IT should be via controlled jump servers with logging and MFA.

b) Strong Identity and Access Management

- Enforce multi-factor authentication for all administrative accounts.
- Implement least privilege – separate engineering, operations, and admin roles.

- Regularly rotate and monitor privileged credentials; implement password vaulting.
- c) ICS-Specific Monitoring and Detection
 - Deploy an ICS-aware IDS/IPS to monitor protocols such as Modbus, DNP3, etc.
 - Trigger alerts on:
 - Unauthorised write operations.
 - Modbus function codes 5, 6, 15, 16 from untrusted hosts.
 - Any ICS communication to/from internet-routed addresses.
- d) Hardening PLCs and Controllers
 - Disable unused ICS services and ports.
 - Enforce vendor firmware updates and patching of web interfaces.
 - Use access control lists (ACLs) to restrict which hosts can talk to PLCs.
- e) Logging, SIEM, and Threat Hunting
 - Centralise logs from:
 - Firewalls and proxies.
 - Servers, domain controllers.
 - SCADA hosts and engineering workstations.
 - Develop specific detection rules for:
 - C2-like beaconing behaviour.
 - Lateral SMB movement from non-standard hosts.
 - Repeated authentication failures on sensitive accounts.

6.3. Counter-Disinformation Measures

Technical defence alone is not enough; the information battle must be fought deliberately.

- a) Pre-Prepared Crisis Communication Playbooks
 - Develop and rehearse a CNI crisis communication plan that includes:
 - Who speaks (named spokespersons).
 - Which channels are used (web, TV, radio, social media).
 - How quickly a first holding statement is released.
- b) Official, Trusted Information Channels
 - Maintain a single authoritative online hub (e.g. gov.uk / Thames Water status portal) where the public can check the real-time status of water safety.
 - Use verified social media accounts with blue ticks and clear branding.
- c) Social Media Monitoring and Rapid Response
 - Implement continuous monitoring for:
 - Keywords like “poisoned water”, “toxic tap”, “Thames Water contamination”.
 - Sudden spikes in mentions across regions.
 - When disinformation is detected:
 - Rapidly publish clear, factual counters.
 - Use infographics and short videos explaining the actual situation.
 - Engage with platforms to flag and remove malicious content.
- d) Media and Public Education
 - Run periodic campaigns explaining:
 - Basics of how water treatment and testing actually work.
 - How to verify official information.
 - The existence of hostile disinformation campaigns targeting utilities.

This reduces the effectiveness of future psychological operations by inoculating the public against fake narratives.

6.4. Coordinated Incident Response Plan

The response to hybrid attacks must be joint between technical, operational, and communication teams.

Phase 1 – Preparation

- Define roles and responsibilities between:
 - Cybersecurity / SOC
 - OT engineering teams
 - Communications / PR
 - Legal and regulatory teams
 - Law enforcement and national cyber centres (e.g. NCSC)
- Conduct joint tabletop exercises focusing on combined cyber + misinformation scenarios.

Phase 2 – Detection and Analysis

- SOC detects suspicious activity (C2, internal scans, ICS anomalies).
- Simultaneously, comms team detects online narratives about “toxic water”.
- Information is correlated to confirm a hybrid attack, not an isolated technical failure or random rumour.

Phase 3 – Containment

- Technical:
 - Isolate compromised hosts.
 - Block outbound connections to known C2 IPs.
 - Restrict access to ICS from potentially compromised segments.
- Communication:
 - Release an immediate, honest “We are investigating, here’s what we know” statement.
 - Warn the public about circulating fake content without amplifying it.

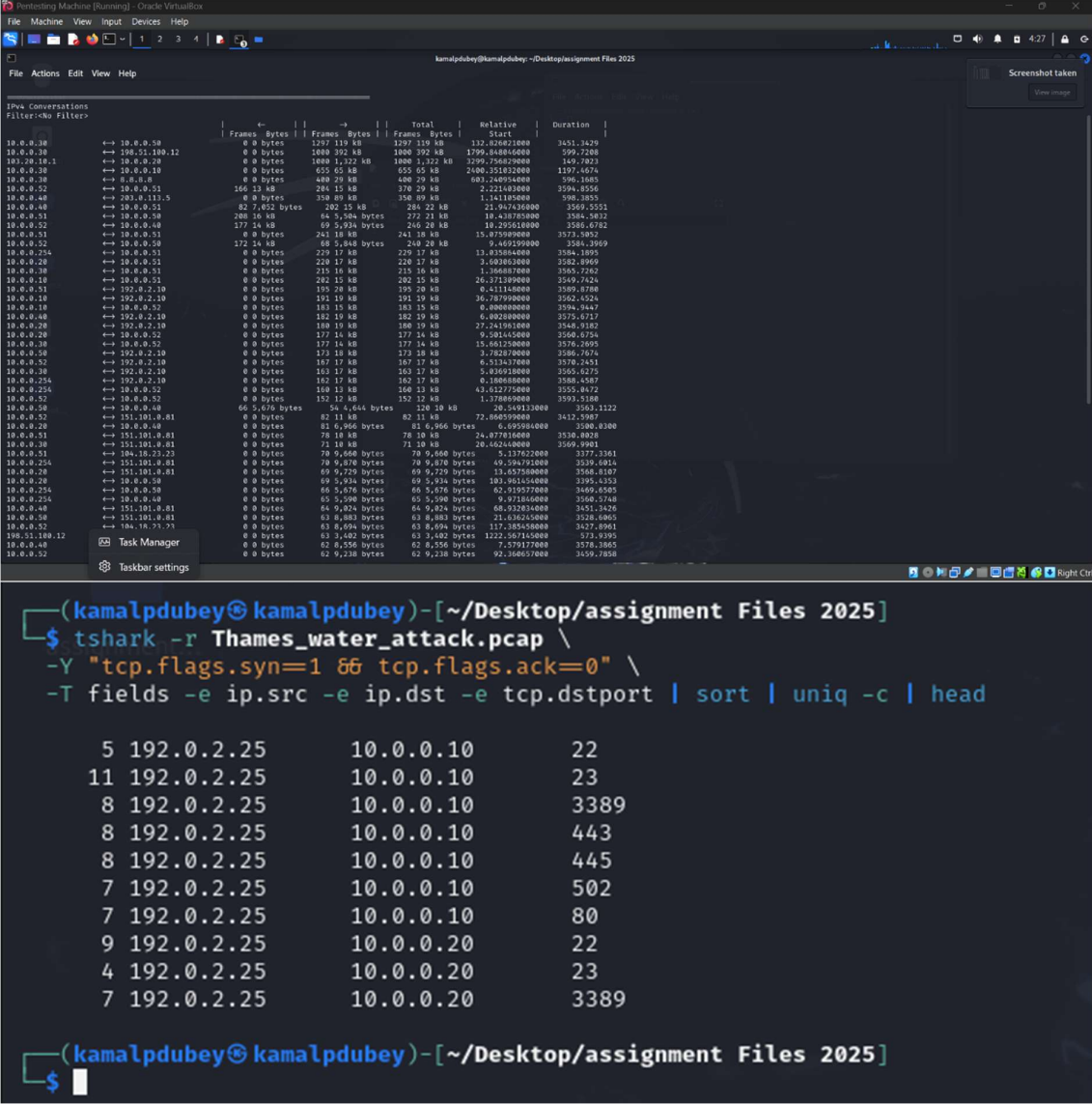
Phase 4 – Eradication and Recovery

- Remove malware and close exploited vulnerabilities.
- Verify all ICS configurations and dosing setpoints against known-good baselines.
- Gradually restore normal connectivity under monitored conditions.
- Continue to counter false narratives until they lose virality.

Phase 5 – Post-Incident Review

- Technical:
 - Forensic report, root cause analysis, and architectural improvements.
- Communication:
 - Review timing and effectiveness of public messaging.
 - Identify which platforms and narratives were most damaging.
- Strategic:
 - Share anonymised threat intelligence with national bodies and sector peers.
 - Update sector-wide guidance on hybrid threats to CNI.

Appendix



The screenshot shows a Kali Linux virtual machine environment. The top window displays a table of IPv6 conversations, and the bottom window shows a terminal with a tshark command and its output.

IPv6 Conversations Table:

Filter: <No Filter>	Frames	Bytes	Frames	Bytes	Total	Relative	Duration
						Start	
10.0.0.30 → 192.0.2.25	0	0 bytes	1297	119 kb	1297	119 kb	3631.2629
10.0.0.30 → 192.0.2.25	0	0 bytes	1000	392 kb	1000	392 kb	399.7208
10.0.0.30 → 192.0.2.25	0	0 bytes	1000	1,322 kb	1000	1,322 kb	2299.756820000
10.0.0.30 → 192.0.2.25	0	0 bytes	655	65 kb	655	65 kb	1197.4676
10.0.0.30 → 192.0.2.25	0	0 bytes	400	29 kb	400	29 kb	603.240954000
10.0.0.30 → 192.0.2.25	166	23 kb	284	15 kb	370	29 kb	3594.8556
10.0.0.30 → 192.0.2.25	0	0 bytes	358	89 kb	358	89 kb	1.141185000
10.0.0.30 → 192.0.2.25	82	7,622 bytes	202	15 kb	284	22 kb	21.947436000
10.0.0.30 → 192.0.2.25	200	16 kb	46	5,304 bytes	272	21 kb	18.438703000
10.0.0.30 → 192.0.2.25	177	14 kb	69	5,924 bytes	246	20 kb	18.295518000
10.0.0.30 → 192.0.2.25	0	0 bytes	241	18 kb	241	18 kb	15.875909000
10.0.0.30 → 192.0.2.25	172	14 kb	68	5,848 bytes	240	20 kb	9.469199000
10.0.0.30 → 192.0.2.25	0	0 bytes	229	17 kb	229	17 kb	13.813884000
10.0.0.30 → 192.0.2.25	0	0 bytes	220	17 kb	220	17 kb	3.620303000
10.0.0.30 → 192.0.2.25	0	0 bytes	215	16 kb	215	16 kb	1.366887000
10.0.0.30 → 192.0.2.25	0	0 bytes	202	15 kb	202	15 kb	26.272309000
10.0.0.30 → 192.0.2.25	0	0 bytes	195	20 kb	195	20 kb	0.411114000
10.0.0.30 → 192.0.2.25	0	0 bytes	191	19 kb	191	19 kb	36.787998000
10.0.0.30 → 192.0.2.25	0	0 bytes	183	15 kb	183	15 kb	6.800000000
10.0.0.30 → 192.0.2.25	0	0 bytes	182	19 kb	182	19 kb	6.002800000
10.0.0.30 → 192.0.2.25	0	0 bytes	180	19 kb	180	19 kb	27.241991000
10.0.0.30 → 192.0.2.25	0	0 bytes	177	14 kb	177	14 kb	9.501650000
10.0.0.30 → 192.0.2.25	0	0 bytes	177	14 kb	177	14 kb	15.061250000
10.0.0.30 → 192.0.2.25	0	0 bytes	173	18 kb	173	18 kb	3.702870000
10.0.0.30 → 192.0.2.25	0	0 bytes	167	17 kb	167	17 kb	6.511617000
10.0.0.30 → 192.0.2.25	0	0 bytes	163	17 kb	163	17 kb	5.830658000
10.0.0.30 → 192.0.2.25	0	0 bytes	162	17 kb	162	17 kb	0.108658000
10.0.0.30 → 192.0.2.25	0	0 bytes	160	15 kb	160	15 kb	43.612775000
10.0.0.30 → 192.0.2.25	0	0 bytes	152	12 kb	152	12 kb	1.378699000
10.0.0.30 → 192.0.2.25	66	5,676 bytes	54	4,644 bytes	120	10 kb	20.549113000
10.0.0.30 → 192.0.2.25	82	11 kb	82	11 kb	72	6,065,990,000	3412.5987
10.0.0.30 → 192.0.2.25	0	0 bytes	81	6,966 bytes	81	6,966 bytes	6.595950000
10.0.0.30 → 192.0.2.25	78	18 kb	78	18 kb	78	18 kb	24.877616000
10.0.0.30 → 192.0.2.25	0	0 bytes	71	10 kb	71	10 kb	20.462400000
10.0.0.30 → 192.0.2.25	0	0 bytes	70	9,660 bytes	70	9,660 bytes	5.137620000
10.0.0.30 → 192.0.2.25	0	0 bytes	70	9,670 bytes	70	9,670 bytes	45.584725000
10.0.0.30 → 192.0.2.25	0	0 bytes	69	9,729 bytes	69	9,729 bytes	13.657580000
10.0.0.30 → 192.0.2.25	0	0 bytes	69	5,924 bytes	69	5,924 bytes	102.961654000
10.0.0.30 → 192.0.2.25	0	0 bytes	66	5,676 bytes	66	5,676 bytes	62.919577000
10.0.0.30 → 192.0.2.25	0	0 bytes	65	5,590 bytes	65	5,590 bytes	9.971840000
10.0.0.30 → 192.0.2.25	0	0 bytes	64	5,924 bytes	64	5,924 bytes	60.532034000
10.0.0.30 → 192.0.2.25	0	0 bytes	63	8,883 bytes	63	8,883 bytes	21.636215000
10.0.0.30 → 192.0.2.25	0	0 bytes	63	6,096 bytes	63	6,096 bytes	117.108565000
10.0.0.30 → 192.0.2.25	0	0 bytes	63	3,482 bytes	63	3,482 bytes	1222.587150000
10.0.0.30 → 192.0.2.25	0	0 bytes	62	6,556 bytes	62	6,556 bytes	7.570177000
10.0.0.30 → 192.0.2.25	0	0 bytes	62	9,718 bytes	62	9,718 bytes	36.360557000

Terminal Output:

```
(kamalpdubey@kamalpdubey)~[~/Desktop/assignment Files 2025]
$ tshark -r Thames_water_attack.pcap \
-Y "tcp.flags.syn==1 && tcp.flags.ack==0" \
-T fields -e ip.src -e ip.dst -e tcp.dstport | sort | uniq -c | head

5 192.0.2.25 10.0.0.10 22
11 192.0.2.25 10.0.0.10 23
8 192.0.2.25 10.0.0.10 3389
8 192.0.2.25 10.0.0.10 443
8 192.0.2.25 10.0.0.10 445
7 192.0.2.25 10.0.0.10 502
7 192.0.2.25 10.0.0.10 80
9 192.0.2.25 10.0.0.20 22
4 192.0.2.25 10.0.0.20 23
7 192.0.2.25 10.0.0.20 3389
```



```
File Machine View Input Devices Help
kamal@dubey@kamal@dubey: ~/Desktop/assignment Files 2025
File Actions Edit View Help
7 192.0.2.25 10.0.0.10 80
9 192.0.2.25 10.0.0.20 22
4 192.0.2.25 10.0.0.20 23
7 192.0.2.25 10.0.0.20 3389

[kamal@dubey@kamal@dubey: ~/Desktop/assignment Files 2025]
# tshark -r 'Thames_water_attack.pcap' \
-v 'tcp.dstport==502 || tcp.dstport==44818 || tcp.dstport==20000' \
-f 'fields == frame.time -> ip.src -> ip.dst -> tcp.dstport'

Oct 27, 2025 09:20:04.305692000 GMT 192.0.2.25 10.0.0.10 502
Oct 27, 2025 09:20:19.814026000 GMT 203.0.113.5 10.0.0.50 502
Oct 27, 2025 09:20:20.846697000 GMT 192.0.2.25 10.0.0.20 502
Oct 27, 2025 09:20:39.709500000 GMT 198.51.100.12 10.0.0.20 502
Oct 27, 2025 09:20:54.027451000 GMT 203.0.113.5 10.0.0.20 502
Oct 27, 2025 09:20:50.322794000 GMT 198.51.100.12 10.0.0.50 502
Oct 27, 2025 09:21:27.802033000 GMT 198.51.100.12 10.0.0.20 502
Oct 27, 2025 09:21:29.191932000 GMT 192.0.2.25 10.0.0.10 502
Oct 27, 2025 09:21:31.981150000 GMT 203.0.113.5 10.0.0.50 502
Oct 27, 2025 09:21:31.329050000 GMT 198.51.100.12 10.0.0.50 502
Oct 27, 2025 09:21:32.631570000 GMT 198.51.100.12 10.0.0.10 502
Oct 27, 2025 09:21:42.499050000 GMT 198.51.100.12 10.0.0.50 502
Oct 27, 2025 09:21:40.251000000 GMT 203.0.113.5 10.0.0.50 502
Oct 27, 2025 09:21:50.301751000 GMT 192.0.2.25 10.0.0.20 502
Oct 27, 2025 09:21:53.875200000 GMT 203.0.113.5 10.0.0.20 502
Oct 27, 2025 09:22:00.651587000 GMT 192.0.2.25 10.0.0.20 502
Oct 27, 2025 09:22:11.846500000 GMT 198.51.100.12 10.0.0.10 502
Oct 27, 2025 09:22:22.222320000 GMT 198.51.100.12 10.0.0.20 502
Oct 27, 2025 09:22:24.553600000 GMT 192.0.2.25 10.0.0.20 502
Oct 27, 2025 09:22:25.697230000 GMT 198.51.100.12 10.0.0.50 502
Oct 27, 2025 09:22:44.963591000 GMT 198.51.100.12 10.0.0.50 502
Oct 27, 2025 09:22:44.977670000 GMT 203.0.113.5 10.0.0.50 502
Oct 27, 2025 09:22:57.920647000 GMT 203.0.113.5 10.0.0.20 502
Oct 27, 2025 09:23:10.260550000 GMT 198.51.100.12 10.0.0.10 502
Oct 27, 2025 09:23:20.837054000 GMT 198.51.100.12 10.0.0.50 502
Oct 27, 2025 09:23:22.679200000 GMT 198.51.100.12 10.0.0.20 502
Oct 27, 2025 09:23:45.754416000 GMT 198.51.100.12 10.0.0.20 502
Oct 27, 2025 09:23:41.306000000 GMT 203.0.113.5 10.0.0.50 502
Oct 27, 2025 09:23:47.546024000 GMT 203.0.113.5 10.0.0.10 502
Oct 27, 2025 09:24:29.340077000 GMT 203.0.113.5 10.0.0.10 502
Oct 27, 2025 09:24:31.853700000 GMT 192.0.2.25 10.0.0.10 502
Oct 27, 2025 09:24:34.172520000 GMT 192.0.2.25 10.0.0.50 502
Oct 27, 2025 09:24:35.661640000 GMT 198.51.100.12 10.0.0.20 502
Oct 27, 2025 09:24:39.406512000 GMT 192.0.2.25 10.0.0.10 502
Oct 27, 2025 09:24:42.257230000 GMT 192.0.2.25 10.0.0.20 502
Oct 27, 2025 09:25:01.901941000 GMT 203.0.113.5 10.0.0.20 502
Oct 27, 2025 09:25:09.373910000 GMT 203.0.113.5 10.0.0.50 502
Oct 27, 2025 09:25:14.564057000 GMT 203.0.113.5 10.0.0.20 502
Oct 27, 2025 09:25:15.000054000 GMT 192.0.2.25 10.0.0.20 502
Oct 27, 2025 09:25:16.727050000 GMT 203.0.113.5 10.0.0.20 502
Oct 27, 2025 09:25:25.636070000 GMT 192.0.2.25 10.0.0.20 502
Oct 27, 2025 09:25:41.952397000 GMT 192.0.2.25 10.0.0.50 502
```

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File Machine View Input Devices Help
kamal@dubey@kamal@dubey: ~/Desktop/assignment Files 2025
File Actions Edit View Help
444 20.100.176.5
440 0.0.0.0
163 192.0.2.25

[kamal@dubey@kamal@dubey: ~/Desktop/assignment Files 2025]
# tshark -r 'Thames_water_attack.pcap' \
-v 'ip.addr==203.0.113.5' \
-f 'fields == frame.time_relative -> ip.src -> ip.dst -> tcp.len | head -n 40'

1.141105000 10.0.0.40 203.0.113.5 295
0.043300000 10.0.0.40 203.0.113.5 293
6.411123000 10.0.0.40 203.0.113.5 298
6.974940000 10.0.0.40 203.0.113.5 295
7.114960000 10.0.0.40 203.0.113.5 293
16.004621000 10.0.0.40 203.0.113.5 293
21.772504000 10.0.0.40 203.0.113.5 295
24.223340000 10.0.0.40 203.0.113.5 293
29.927627000 10.0.0.40 203.0.113.5 295
37.709064000 10.0.0.40 203.0.113.5 295
36.198420000 10.0.0.40 203.0.113.5 298
39.772014000 10.0.0.40 203.0.113.5 295
61.924025000 10.0.0.40 203.0.113.5 295
61.903000000 10.0.0.40 203.0.113.5 298
62.321704000 10.0.0.40 203.0.113.5 298
62.040755000 10.0.0.40 203.0.113.5 298
65.472120000 10.0.0.40 203.0.113.5 295
66.019167000 10.0.0.40 203.0.113.5 295
69.200003000 10.0.0.40 203.0.113.5 292
69.030792000 10.0.0.40 203.0.113.5 295
66.030562000 10.0.0.40 203.0.113.5 295
55.445167000 10.0.0.40 203.0.113.5 298
55.662050000 10.0.0.40 203.0.113.5 295
59.093222000 10.0.0.40 203.0.113.5 298
59.453330000 10.0.0.40 203.0.113.5 295
63.051300000 10.0.0.40 203.0.113.5 293
65.630020000 10.0.0.40 203.0.113.5 298
65.310047000 10.0.0.40 203.0.113.5 298
68.037602000 10.0.0.40 203.0.113.5 295
69.599160000 10.0.0.40 203.0.113.5 298
72.076440000 10.0.0.40 203.0.113.5 295
74.379270000 10.0.0.40 203.0.113.5 295
76.632990000 10.0.0.40 203.0.113.5 298
78.110047000 10.0.0.40 203.0.113.5 293
79.039010000 10.0.0.40 203.0.113.5 298
80.155244000 10.0.0.40 203.0.113.5 293
80.700411000 10.0.0.40 203.0.113.5 295
81.152100000 10.0.0.40 203.0.113.5 293
81.030211000 10.0.0.40 203.0.113.5 298
89.229182000 10.0.0.40 203.0.113.5 295

[kamal@dubey@kamal@dubey: ~/Desktop/assignment Files 2025]
```

